

AI in Marketing Capstone

Concept Note and Implementation Plan Submission Guidelines

Project Information

Project Title: OnBrand AdCopy

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Concept Note

1. Project Overview

- Digital marketing and advertising teams are under continuous pressure to create, test, and iterate dozens of ad copy variants across multiple digital channels (Instagram, Twitter, Facebook, Reddit). Writing these variants manually is time-consuming and labor-intensive, and marketing teams face great uncertainty regarding which copy variant will yield higher click-through rates (CTR) or conversions prior to live deployment. Simultaneously, performance-driven copywriting frequently degrades brand identity by adopting generic, clickbait, or aggressive urgency-focused language that violates established brand voice guidelines. The OnBrand AdCopy project addresses this critical trade-off by offering an integrated, AI-driven decision support system that generates multi-style ad copy variants, predicts their CTR performance using machine learning, evaluates their semantic alignment with brand voice guidelines using sentence embeddings, and automatically rewrites off-brand variants through an automated feedback loop.
- The marketing context focuses on multi-channel digital advertising across social media and search platforms targeting consumer audiences at the awareness and consideration stages of the customer journey. The system is designed for marketing professionals, brand managers, and digital strategists managing campaigns for enterprise consumer brands. The project empirically demonstrates performance across three distinct brand voice profiles: Samsung (innovative, empowering, premium, tech-forward), Duolingo (witty, playful, pushy/nagging, casual, gamified), and Nike (action-oriented, inspiring, bold, athletic empowerment). The primary business processes involved are campaign creative generation, quality assurance, tonal audit, and pre-campaign creative performance optimization.
- Solving this dual challenge provides substantial customer and business impact. From a business perspective, automating copy generation and pre-launch quality scoring reduces creative production cycles from days to second while eliminating costly A/B testing of low-performing or brand-damaging copy variants. McKinsey Global Institute (2023) estimates that generative AI could add \$400-660 billion annually in value to marketing and sales functions, with creative copy optimization being a leading driver. BCG (2023) reports that while 70% of CMOs deploy generative AI, brand safety and consistency remain their top operational risks. OnBrand AdCopy delivers measurable business value by enabling marketing teams to scale content production without diluting brand equity or compromising predicted engagement metrics.

2. Objectives and Marketing KPIs

- The primary objective of the OnBrand AdCopy project is to develop and validate an end-to-end AI platform that automates the generation, evaluation, and optimization of digital ad copy. Specifically, the project aims to: (1) engineer an LLM prompting pipeline to generate candidate copy variants across 4 distinct styles (emotional, informative, urgency-focused, humorous); (2) build an XGBoost predictive model to estimate Click-Through Rate (CTR proxy) from structural and NLP text features; (3) implement a Sentence-BERT (SBERT) cosine similarity module to score brand voice alignment against approved reference corpora; (4) combine these outputs into a configurable composite score; and (5) establish an automated LLM auto-correction loop that rewrites off-brand variants to satisfy brand alignment thresholds while preserving performance.

- The project directly targets two core marketing KPIs: (1) Click-Through Rate (CTR Proxy %), calculated as total engagements (likes + shares + comments) divided by impressions, serving as the primary proxy for creative engagement; and (2) Brand Voice Alignment Score (0-100%), measuring the semantic cosine similarity between generated copy and brand reference benchmarks. Secondary evaluation metrics include text fluency, stylistic distinctiveness across brand profiles, and production cycle speed-up.
- The quantitative success criteria and targets established for the project are: (1) CTR Prediction Accuracy: achieve an R-squared > 0.15 / RMSE < 0.25 on held-out test data for the tabular XGBoost model, outperforming baseline linear regression models; (2) Brand Alignment Threshold: maintain an average Sentence-BERT cosine similarity score ≥ 0.75 (75%) across generated copy for all three brand profiles (Samsung, Duolingo, Nike); (3) Auto-Correction Efficiency: successfully uplift low-alignment variants ($< 70\%$ alignment) to $\geq 75\%$ alignment following LLM rewriting while experiencing $< 5\%$ drop in predicted CTR score; and (4) Composite Score Optimization: demonstrate that top-ranked composite variants outperform unconstrained LLM baseline copy in combined engagement and brand fit metrics.

3. Background and Marketing Context

- In modern performance marketing, ad copy is the primary touchpoint between a brand and potential customers on digital feeds. With ad fatigue accelerating and attention spans declining, campaign success depends on rapid creative iteration. However, current industry practice relies either on manual copywriting which cannot scale to meet multi-channel demands or unconstrained generative AI tools (such as generic ChatGPT prompts) that produce bland, homogeneous, or off-brand copy. Brand managers are currently forced to conduct manual review using static brand guidelines books, a process that is slow, subjective, unscalable, and unquantifiable.
- Existing marketing technology solutions typically operate in silos: copy generation platforms (e.g., Copy.ai, Jasper) focus purely on text generation without predictive performance feedback or rigorous brand scoring; conversely, tech analytics platforms evaluate historical campaign metrics post-launch without offering pre-launch creative correction. Furthermore, standard text style transfer algorithms in NLP research focus on coarse binary dimensions (formal/informal or positive/negative sentiment) rather than nuanced, multi-faceted enterprise brand voice profiles.
- An AI-based approach adds transformative value by unifying generation, predictive analytics, and semantic alignment into a single closed-loop system. Machine learning algorithms (XGBoost) capture complex, non-linear interactions between text length, punctuation, sentiment, and engagement that human copywriters cannot systematically calculate. Sentence-BERT embeddings convert qualitative brand voice rules into precise, multi-dimensional vector comparisons. Finally, LLMs provide the generative flexibility to re-synthesize text based on real-time mathematical feedback. This hybrid AI architecture converts brand compliance from a subjective bottleneck into a real-time, automated performance engine.

4. Proposed AI Methodology

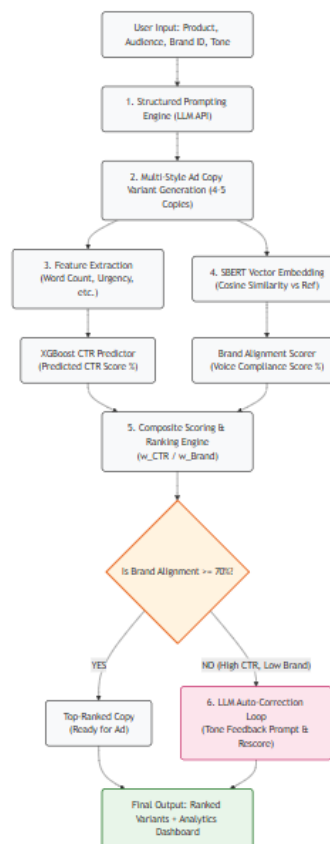
- The proposed OnBrand AdCopy methodology follows a 6-stage sequential AI pipeline: (1) Structured Input & Prompting: user inputs product description, target audience, brand ID, and desired tone parameters into a structured prompt template; (2) Multi-Style LLM Generation: an LLM API generates 4-5 copy variants spanning emotional, informative, urgency, and humorous styles; (3) Feature Extraction & XGBoost CTR Prediction: text features (char count, word count, question mark, exclamation mark, urgency count, sentiment score) are extracted and passed to a pre-trained XGBoost model to yield an estimated CTR score; (4) SBERT Semantic Alignment Scoring: Sentence-BERT (all-MiniLM-L6-v2) embeds the generated copy and computes cosine similarity against the brand's reference corpus, combined with rule-based style checks (formality, vocabulary constraints); (5) Composite Weighting & Ranking: variants are ranked using a user-configurable composite score $S_{comp} = w * S_{CTR} + (1 - w) * S_{brand}$ (e.g., 60% performance / 40% brand alignment);

and (6) AutoCorrection Loop: variants exceeding CTR thresholds but failing brand alignment (<70%) are automatically passed back to the LLM with specific alignment feedback for rewriting and re-scoring.

- The primary technical models and tools utilized in this architecture include: (1) Generative LLMs (OpenAI GPT-4o / Claude 3.5 Sonnet API) for structured variant generation and targeted rewriting; (2) XGBoost Regressor (scikit-learn / xgboost library) trained on Kaggle social media engagement data for non-linear CTR regression; (3) Sentence-Transformers (Sentence-BERT all-MiniLM-L6-v2) for 384-dimensional dense semantic embedding and vector similarity calculation; and (4) Python NLP libraries (pandas, numpy, re, nltk/spacy) for text normalization, regex feature extraction, and statistical pre-processing.
- This hybrid approach is specifically appropriate for the project because: (1) tabular ensemble methods (XGBoost) consistently outperform deep neural networks on structured feature datasets of moderate size (~1,200 records), avoiding overfitting while providing feature importance interpretability; (2) SBERT vector embeddings capture deep semantic context and brand tone without requiring expensive fine tuning of whole language models; (3) API-based LLMs provide state-of-the-art zero-shot and few-shot text generation capabilities; and (4) the composite score grants brand managers explicit control over the trade-off between aggressive marketing performance and strict brand compliance.
- The solution will be evaluated using a comprehensive dual framework: (1) Technical Evaluation: 5-fold cross-validation on the CTR dataset evaluating RMSE, MAE, and R-squared against baseline models (Linear Regression, Random Forest), alongside SBERT cosine similarity validation across on-brand vs. out-of-brand text pairs; and (2) Marketing/Business Evaluation: comparative analysis of generated variants against baseline unconstrained copy, evaluating alignment uplift (percentage increase in SBERT score after autocorrection) and trade-off efficiency (retained CTR percentage post-correction).

5. Architecture / Workflow Design Diagram

- The high-level system architecture of OnBrand AdCopy follows an integrated closed-loop workflow from user input to optimized marketing copy deployment:



- **Component Roles:** (1) Structured Prompting Engine parses user parameters into contextual prompts; (2) Multi-Style Generator produces diverse creative options; (3) Feature Extraction & XGBoost Model computes predicted engagement metrics; (4) SBERT Scorer measures semantic similarity against brand reference JSON profiles; (5) Composite Ranking Engine calculates weighted final scores; (6) Auto-Correction Loop intercepts high-potential off-brand copy and prompts the LLM to rewrite it to meet brand standards; and (7) Analytics Dashboard presents ranked copy options with clear score breakdowns for human review.

6. Data Sources

- The project utilizes two primary data sources: (1) Kaggle Social Media Engagement Dataset: a public dataset containing 12,000 social media posts across 28 features (platform, text_content, impressions, likes_count, shares_count, comments_count, sentiment_score, brand_name, campaign_phase). After filtering for English posts (n=1,197) and non-zero impressions, an engagement CTR proxy metric is derived: $((\text{likes} + \text{shares} + \text{comments}) / \text{impressions}) * 100$, with extreme outliers clipped at the 99th percentile. NLP feature engineering extracts char_count, word_count, has_question, has_exclamation, and urgency_count; and (2) Brand Reference JSON (brand_reference.json): a first-party structured corpus containing verified brand voice summaries, style rules, tone categories, and 15 reference ad copies for Samsung, Duolingo, and Nike. The Kaggle dataset is synthetic/public data requiring no PII or IRB consent; the brand reference JSON contains publicly available commercial copy.

7. Literature and Industry Review

- The project builds upon a robust foundation of 18 academic papers and industry reports. Key technical benchmarks include Chai et al. (Microsoft FAST, 2022) on feedback-aware self-training for controllable text generation; GCOF (2024) on self-iterative LLM copywriting; Bai et al. (2025) on advertising CTR prediction algorithms; Lin et al. (ClickPrompt, 2024) on coupling CTR models with LLM prompt generation; Hassan et al. (2023) on brand personality word embeddings; and Maulina & Hendrawan (2026) on XGBoost feature selection for social media promotion. Industry reports (McKinsey, 2023; BCG, 2023; State of AI in Marketing 2026) confirm high adoption of GenAI in marketing alongside critical concerns regarding brand safety and quality control. The identified gap is that existing literature treats generation, CTR prediction, and style transfer in isolation; OnBrand AdCopy bridges this gap by unifying these components into a closed-loop composite scoring and auto-correction platform.

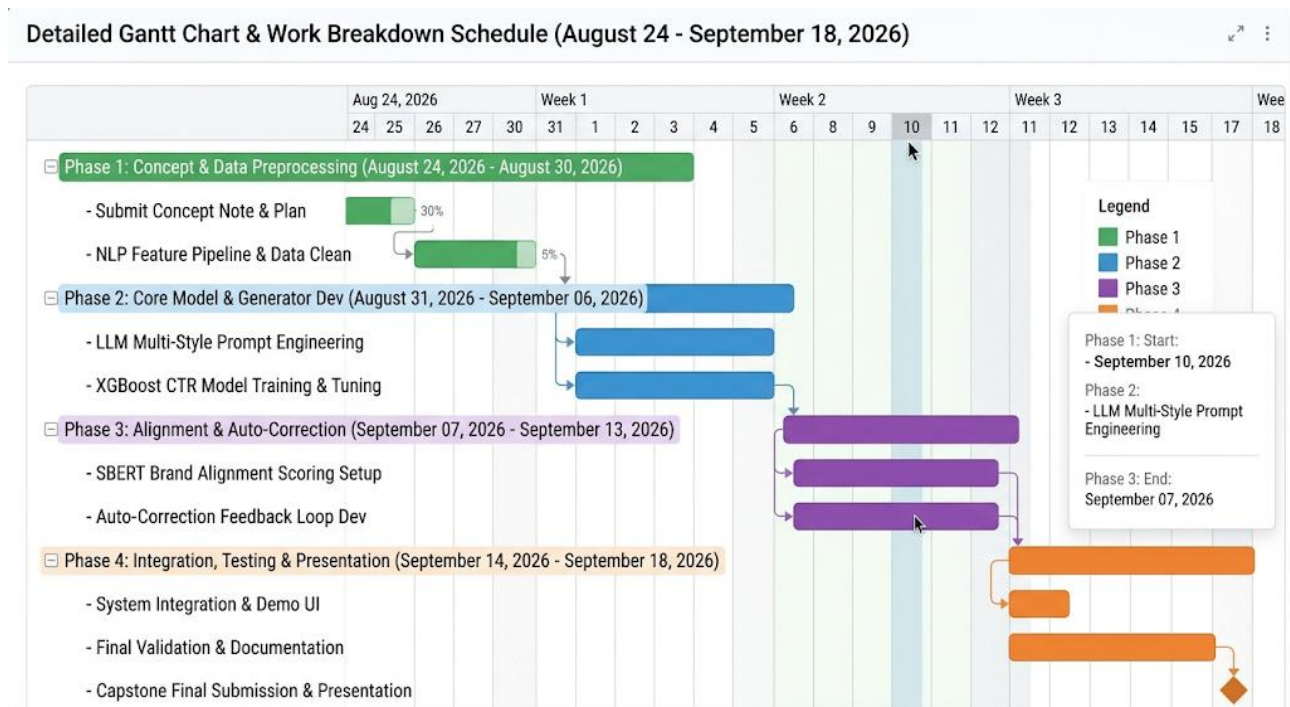
Implementation Plan

1. Technology Stack

- The implementation plan relies on a robust, modern Python data science and AI tech stack organized into four functional layers:
 - **Development Environment & Languages:** Python 3.11+, Jupyter Notebooks (SIC_Capstone_2.ipynb), VS Code, Git/GitHub repository.
 - **AI/ML Libraries & Frameworks:** scikit-learn, XGBoost, Sentence-Transformers (all-MiniLM-L6-v2), PyTorch, NLTK/SpaCy, Regex (re).
 - **Generative LLM APIs:** OpenAI API (GPT-4o / GPT-3.5-turbo) or Anthropic Claude API or Gemini API for structured text generation and rewriting.
 - **Data Processing, Visualization & Deployment:** pandas, numpy, matplotlib, seaborn, Streamlit/Gradio (for interactive demo), Docker.

2. Timeline and Task Distribution

- The capstone implementation schedule aligns directly with the official Samsung Innovation Campus AI in Marketing course timeline (July - September 2026), with final capstone project submission and presentation scheduled for September 18, 2026.



- Task Distribution Matrix:** As an individual capstone project prepared by Tunahan Sanal, all responsibilities across data engineering, machine learning modeling, LLM prompt engineering, SBERT integration, software architecture, testing, and presentation preparation are executed solely by the lead author.

3. Milestones and Deliverables

- Key milestones and concrete evidence of completion for the OnBrand AdCopy implementation include:
 - Milestone 1: Data Preprocessing & Feature Engineering Complete (Aug 30, 2026)**
Evidence: Cleaned English dataset (n=1,197) with extracted NLP features (word count, urgency, sentiment) and validated CTR proxy metric.
 - Milestone 2: XGBoost CTR Model Trained & Validated (Sep 05, 2026)**
Evidence: Trained XGBoost model achieving $R^2 > 0.15$ on test split, with feature importance plots and saved model artifact (.json/.pkl).
 - Milestone 3: SBERT Brand Alignment & Multi-Style Generator Operational (Sep 09, 2026)**
Evidence: Working SBERT embedding pipeline calculating cosine similarity against brand_reference.json profiles for Samsung, Duolingo, Nike.
 - Milestone 4: Closed-Loop AutoCorrection Pipeline Integrated (Sep 12, 2026)**
Evidence: Functional Python pipeline executing variant generation -> XGBoost scoring -> SBERT scoring -> composite ranking -> autocorrection rewriting.
 - Milestone 5: Interactive Demo UI & Final Presentation Ready (Sep 18, 2026)**
Evidence: Working Streamlit/Gradio web interface, complete project repository, walkthrough documentation, and final presentation slide deck for SIC Capstone Day.

4. Challenges and Mitigation Strategies

- The project identifies four major technical and operational risks along with clear mitigation strategies and fallback options:
 - **Risk 1:** Small / Synthetic Training Dataset (n=1,197 English posts) leading to potential model overfitting. **Mitigation:** Use 5-fold cross-validation, strong L1/L2 regularization in XGBoost, and hyperparameter tuning. **Fallback:** Supplement feature set with rule-based heuristics if regression variance is high.
 - **Risk 2:** LLM Hallucination and Brand Safety Violations during text generation or rewriting. **Mitigation:** Implement strict system prompts with explicit negative constraints (e.g., forbidden words, price claims) and post-generation regex filters. **Fallback:** Human-in-the-loop approval button in the UI before copy export.
 - **Risk 3:** AutoCorrection Trade-Off (rewriting for brand tone reduces predicted CTR). **Mitigation:** Track CTR delta post-rewriting; if CTR drops >10%, present both the original high-CTR variant and the corrected on-brand variant to the user with explicit score trade-offs. **Fallback:** User-adjustable composite weight slider (w_CTR vs. w_Brand).
 - **Risk 4:** API Rate Limits, Latency, and Cost for LLM calls. **Mitigation:** Cache generated responses, use efficient smaller models (GPT-4o-mini or Claude 3.5 Haiku) for initial variant generation, and reserve full models for autocorrection. **Fallback:** Pre-generated offline variant demonstration mode.

5. Ethical and Responsible AI Considerations

- **Responsible AI practices are embedded throughout the OnBrand AdCopy architecture:** (1) Transparency & Explainability: the system avoids black-box recommendations by providing explicit sub-scores (predicted CTR %, SBERT brand alignment %, feature importance drivers) for every generated variant; (2) Prevention of Manipulative Copywriting: system prompts prohibit deceptive claims, false scarcity, or toxic urgency patterns, directly informed by our EDA finding that aggressive urgency keywords correlate negatively with engagement; (3) Brand Safety & Moderation: automated toxicity filtering (using dataset toxicity thresholds and LLM moderation APIs) prevents generation of harmful or offensive language; and (4) Human Oversight: the platform functions strictly as a decision-support tool, leaving final campaign publishing decisions to human marketing managers.
- **AI Disclosure Notice:** This document was authored by Tunahan Şanal for the Samsung Innovation Campus AI in Marketing Capstone Program, utilizing AI assistance for research synthesis, document formatting, and text refinement.

6. References

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